

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1410

Baytown Energy Center 500 kV
August 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1410. The Interconnection Customer proposes a 500.0 MW solar plus storage facility interconnecting at the Baytown Energy Center 500 kV substation in the Illinois area. The request was received on February 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 447 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,429,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PHELPS DODGE / EASY RD 230 KV	PHELPS DODGE / NRG ROCKFORD I 230 KV	101.9	625	10.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PHELPS DODGE / EASY RD 230 KV	Reconductor line	\$22,449,000	24% (\$5,429,000)
	Total allocated to project		\$5,429,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.